Cobordism Distance

Knot Theory Research

November 13, 2026

Abstract

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1 Table

For $K, J \subset S^3$, We define the cobordism distance as $d_C(K, J) = g_4(K\# - J)$. This collapses to a metric in $\mathcal{C}$

$d_C(K,J)$	O	3_1	4_1	5_1	5_2
O	0	1	1	2	1
3_1	.	0	x	1	x
4_1	.	.	0	x	x
5_1	.	.	.	0	x
5_2	.	.	.	.	0

Table 1: Concordance distance $d_C(K, J)$ for prime knots up to 5 crossings.

1.1 Table

2 Identities

We have:

$$g_4(K) = g_4(K^r) \tag{1}$$

$$d(K, J) = g_4(\bar{K}, \bar{J}) \tag{2}$$

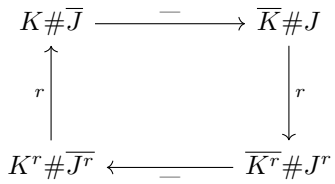
$$d(K^r, J^r) = d(K, J) \tag{3}$$

$$d(K^r, J) = d(K, J^r) \tag{4}$$

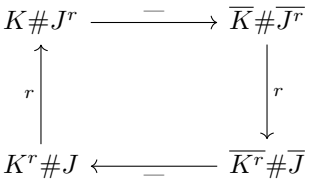
1. Equivalence Class: $K\#J$

$$\begin{array}{ccc} K\#J & \xrightarrow{\quad - \quad} & \overline{K}\#\overline{J} \\ {\scriptstyle r}\uparrow & & \downarrow{\scriptstyle r} \\ K^r\#J^r & \xleftarrow[\quad - \quad]{} & \overline{K^r}\#\overline{J^r} \end{array}$$

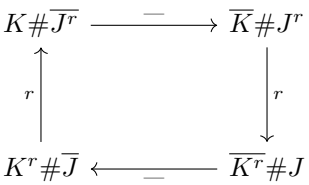
2. Equivalence Class: $K\#\overline{J}$



3. Equivalence Class: $K\#J^r$



4. Equivalence Class: $K\#\overline{J^r}$



3 A Lower Bound

We know that the Cobordism Distance is subadditive, i.e. $g_4(K\#J) \leq g_4(K) + g_4(J)$. From here we can show

$$\begin{aligned} g_4(K) &= g_4(K\#\overline{J}\#J) \\ \implies g_4(K) &\leq g_4(K\#\overline{J}) + g_4(J) \\ \implies g_4(K) - g_4(J) &\leq g_4(K\#\overline{J}) \quad \blacksquare \end{aligned}$$

Similarly,

$$\begin{aligned} g_4(J) &= g_4(J\#\overline{K}\#K) \\ \implies g_4(J) &\leq g_4(J\#\overline{K}) + g_4(K) \\ \implies g_4(J) - g_4(K) &\leq g_4(J\#\overline{K}) \\ &= g_4(\overline{J\#\overline{K}}) \\ &= g_4(\overline{J}\#K) \\ &= g_4(K\#\overline{J}) \quad \blacksquare \end{aligned}$$

Thus,

$$|\, g_4(K) - g_4(J)\,| \leq g_4(K\#\overline{J}) \leq g_4(K) + g_4(J) \quad \square$$